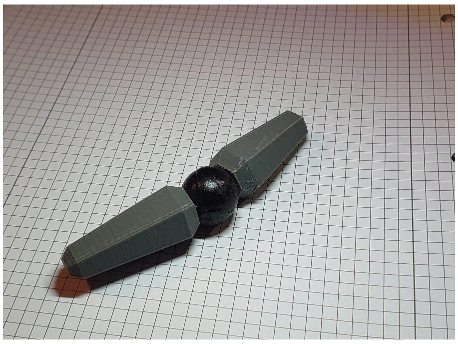
MicroGrid swarm-based, modular mini robots

Preliminary Design Review
WRO 2026, Hungary, Regionals

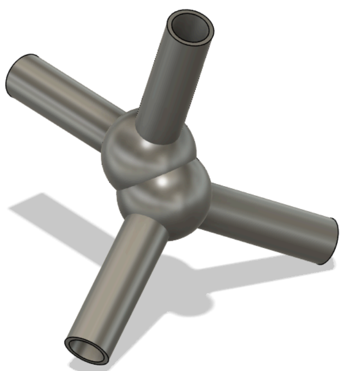
Team Members

Péter Szita - Engineer, responsible for the robot structural and movement design and 3D printing, and outreach in his area.
Nándor Sümeghi - Project manager, programmer, “businessman”, responsible for the simulation, robot code, theoretical background and outreach in his area.
Mariann Sásdi - Coach, providing guidance and support throughout the project.

Summary



Our goal is to develop a multi-purpose robot swarm that can support various cultural applications. The system will utilize swarm intelligence (SI), a subset of artificial intelligence, along with swarm movement and other swarm robotics concepts.
Each robot will be designed to support both its own weight and that of other robots, as well as additional structures built on top. The robots will be capable of connecting to one another, moving independently, and forming complex structures.
We reached out to professionals, including Tamás Vicsek, an expert in swarm movement, who was open to share his knowledge with us, and Varinex Kft., a 3D printing company. For the national competition, we plan to develop an approximately 15 cm prototype, along with functional code and multiple simulations.



Project Overview

Theme

Our project explores how robotics can enhance culture, art, and creativity. We structured our approach around the three main areas defined in the Future Innovators guide.

Protecting, Preserving & Sharing Cultural Heritage

The swarm’s structure-forming capability allows historical sites to be explored more safely and even partially reconstructed. The robots could support fragile structures, stabilize elements such as rocks or pillars, and enhance accessibility.
Additionally, they could recreate statues, buildings, or historical environments in different locations, making cultural heritage more widely accessible.

Co-Creation: Humans, Robots & AI

The robots can also collaborate with artists, serving as a powerful tool for creative exploration. Using a digital pheromone system, the swarm could be directed collectively rather than controlling each robot individually.
This would allow robots to wrap around objects, build structures, and bring artistic ideas to life more efficiently.

Experiencing Art and History with Robots

Since the robots can form a wide variety of shapes, they can be used to dynamically recreate exhibits and historical scenes. They could restore statues to their original form or create highly interactive and immersive artistic experiences.

Timeline

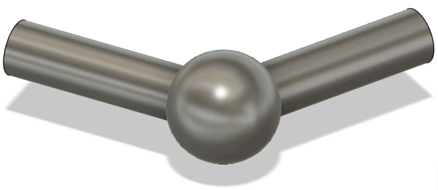
We have already created initial design sketches and conducted research on swarm robotics.

May 1-15.	May 16-31.	June 1-17.	June 18-25.
Structural design Swarm robotics theory, simulation preparation Searching for sponsors and professionals, media promotion	Electrical design Simulation coding and running Searching for sponsors and professionals, media promotion	Finalising 3D design Simulation running, robot coding Media promotion	Reviewing 3D design Reviewing coding and exporting simulations Media promotion

We also plan to make commits on GitHub at larger milestones, this process is still being defined.
For further development, we plan to build a ~5 cm robot for the international round if we qualify. Otherwise, or after this phase, we plan to expand the project at the university level, focusing on smaller and more efficient robots.

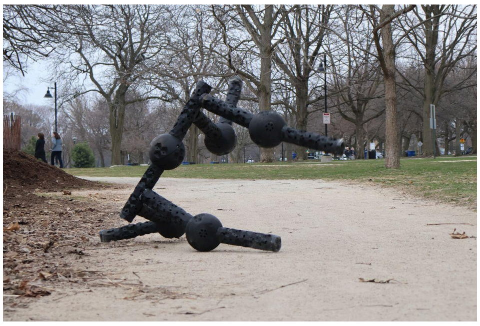
Robot Overview

Design



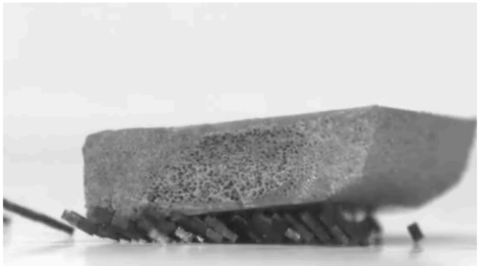
Each robot consists of three main parts: two arms and a central body. The arms will each have two degrees of freedom relative to the body and will be responsible for locomotion, each containing a motor.
The body will house the control electronics. The robots will include multiple connection points to attach to other robots: two at the ends of the arms and three on the body, spaced 120 degrees apart.
The final version will use strong materials such as carbon fiber, with rubber elements for improved grip. The prototype will have similar properties but will be made from plastic, measuring approximately 15 cm in width with a diameter of 3–4 cm.
Planned components include an ESP32 SuperMini, a gyroscope, four motors, and multiple batteries.

Code



The control system will incorporate the Vicsek model to enable coordinated base movement. Modular movement will be based on predefined structural configurations (e.g., tetrahedral structures, lattice wrapping, wheels, helices, telescoping mechanisms, and gears), allowing the swarm to form increasingly complex structures.
The system will also include a pheromone-based guidance mechanism, enabling robots to move toward or away from specific areas. Communication will be achieved through short-range electromagnetic signals, allowing each robot to interact with nearby neighbors.
A gyroscope will assist with movement stabilization and orientation. At initialization, the robots will establish a shared frame of reference to improve coordination.

Key Mechanisms



- Ground-based individual movement
- Modular arrangement into structures
- Pheromone-based guidance
- Lattice wrapping

Sources

AI usage

AI tools were used for grammar correction and rewording in this document, as well as for researching relevant concepts and ideas.

Inspiration

Our project was inspired by several real-world systems and one fictional example:

- Hanyang University, ant-like micro robots: https://www.newscientist.com/article/2461218-swarms-of-tiny-robots-coordinate-to-achieve-ant-like-feats-of-strength/
- Agile Legged Locomotion in Reconfigurable Modular Robots: https://modularlegs.github.io/
- Big Hero 6 “Microbots”: https://www.youtube.com/watch?v=fsVJuN75vzE

Other important sources

Tamás Vicsek’s presentation at Fazekas and email correspondence
https://en.wikipedia.org/wiki/Swarm_robotics
https://hal.elte.hu/drones/
https://en.wikipedia.org/wiki/Vicsek_model
https://en.wikipedia.org/wiki/Swarm_intelligence
https://web.mit.edu/8.334/www/grades/projects/projects10/Hernandez-Lopez-Rogelio/dynamics_2.html
WRO International and Hungarian websites
The pictures are from the inspirational robots or are our designs.